

Model Development Phase Template

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Name

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Project Title

Harnessing EC2, VPC & MobaXterm for Cutting-Edge Entity Hosting

Maximum Marks

5 Marks

Model Selection Report

Component**Description**

Amazon EC2 (Elastic Compute Cloud)

Amazon EC2 is used to create and manage virtual servers in the cloud. It provides scalable computing capacity, allowing users to run applications without investing in physical hardware. EC2 supports multiple instance types based on performance requirements and enables easy deployment of applications. It is cost-effective and flexible but requires proper configuration for security and performance.

Amazon VPC (Virtual Private Cloud)

Amazon VPC provides a secure and isolated network environment within AWS. It allows users to define IP address ranges, create subnets, and configure routing tables and gateways. VPC ensures secure communication between resources and provides control over network access. However, it requires proper setup and understanding of networking concepts.

MobaXterm (Remote Access Tool)

MobaXterm is used for securely connecting to the EC2 instance using SSH. It provides a user-friendly interface for remote server management, file transfer, and command execution. It simplifies remote access but depends on proper key configuration and network connectivity.

Linux Server (Ubuntu)

The Linux operating system is used on the EC2 instance to run and manage server-side operations. It supports various commands and tools for deployment, configuration, and monitoring. It is lightweight and efficient but requires command-line knowledge.

Web Server (Apache/Nginx)

A web server is installed on the EC2 instance to host applications or websites. It handles client requests and delivers web content. It is essential for deployment but needs proper configuration for security and performance.

Conclusion

Final System Selected

EC2 + VPC + MobaXterm + Linux + Web Server

Reasoning

The selected system components work together to provide a complete cloud hosting solution. EC2 offers scalable computing resources, VPC ensures secure networking, and MobaXterm enables efficient remote access. The Linux server and web server allow smooth deployment and management of applications. This combination is reliable, cost-effective, and suitable for real-world cloud infrastructure projects.